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3.4: Research Publications and Awards

3.4.3: Number of research papers published per teacher in the Journals as notified on UGC CARE list during the last five years

Research Journal Paper published in Year Jan-Dec 2023

Sr. No	Research Paper Title	Author Names	Link to Paper
1	Deep Scattering Convolutional Network For Cosmetic Skin Classification	Dr. Dhananjay Kalbande, Shwetambari Borade, Kristen Pereira, Rushil Patel, Sudhanshu Kulkarn	https://ijettjournal.org/Volume-70/Issue-7/IJETT-V70I7P202.pdf
2	Optimization Of Quality Of Service Using Ecaba Protocol In Wireless Body Area Network	Dr. Dhananjay Kalbande, Renuka Pawar	https://www.x-mol.net/paper/article/1613592644995334144
3	Video Analytics Based Multi-Symptoms System For Determining Progression Of Parkinson Disease	Jignesh Sisodia, Dhananjay Kalbande	not available
4	Finding Missing Person Using Ai.	Dr. Nataasha Raul, Madhav Lahoti, Sanchit Gajam, Aditya Kasat,	not available
5	Autoease Android Application For Asd Children	Dr. Kailas Devadkar , Kadambari patangrao , sarika shete , prachi	https://www.irjmets.com/uploadedfiles/paper//issue_1_january_2023/33111/final/fin_irj

		misra	mets1674733265.pdf
6	Video Analytics Based Multi-Symptoms System For Determining Progression Of Parkinson Disease	Jignesh Sisodia, Dhananjay Kalbande	not available
7	Investigating Abdominal Electrocardiogram (ECG) Signals using Scalogram Analysis for Accurate Characterization	Bhagwat Atharav Rajendra, Najib Ghatte	https://drive.google.com/file/d/1WByNHHZrQba-MPdJ2I0VDlnbESIGAECE/view?usp=drive_link
8	Automatic question generation: a review of methodologies, datasets, evaluation metrics, and applications	Nikahat Mulla & Prachi Gharpure	https://link.springer.com/article/10.1007/s13748-023-00295-9
9	Leveraging well-formedness and cognitive level classifiers for automatic question generation on Java technical passages using T5 transformer	Nikahat Mulla & Prachi Gharpure	https://link.springer.com/article/10.1007/s41870-023-01262-2
10	Modelling of FinFET based On bipolar cell of silicon retina	Payal Shah, Reena Sonkusare, Surendra. S. Rathod	https://doi.org/10.1016/j.biosx.2023.100409
11	FinFET based photoreceptor for silicon retina to reduce power consumption	Payal Shah, Reena Sonkusare, Surendra. S. Rathod	https://www.sciencedirect.com/science/article/pii/S2590137023001061?via%3Dihub
12	Development of Software Tool to Identify Liquid Maldistribution in Distillation Columns	Chinmay Walinjkar, Manisha Bansode, Prof. Payal Shah, P. B. Walinjkar	https://ijarsct.co.in/A10088.pdf
13	Interactive Android Application for Marketing and Promoting Small Buisnesses	Mayur Misal,Onkar Tatkare,Nikhita Mangaonkar	https://tojdel.net/journals/tojdel/articles/v11i01c02/v11i01-49.pdf
14	Foodie - Online Food Delivery	Pallavi Thakur,Tirsha	https://docplayer.net/2371609

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15	Intervention Recommendation Web Application Based On User Interests For Mental Well-Being	Manvita Joshi, Chetashri Mahajan, Traividya Korgaonkar, Nataasha Raul, Meghana N	https://ijcrt.org/papers/IJCRT2308476.pdf
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Original Article

Deep Scattering Convolutional Network for Cosmetic Skin Classification

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Abstract - The use of computer-based solutions or artificial intelligence in detecting face skin conditions has progressed considerably over the years. Combining cosmetic skin care concepts with technical advancements has shown noble outputs. In recent times, deep convolutional neural networks have been applied for image classification applications in various domains like healthcare, identification of objects in various aspects, control systems, machine visions, etc., and have shown remarkable results. The scattering network is commonly and consistently utilized on the initial layers of a supervised mixed convolutional model. This study demonstrates that layers in the first layer do not always have to be learned, with pre-defined models attaining the greatest performance to date when competing with Deep CNNs. ResNETs, in particular, have been adopted widely because they can solve the vanishing gradient problem that prevails in most deep networks. So, they have exploited the capacity of wavelet transforms, Scattering Networks, and the Residual Networks model for image classification to identify cosmetic skin problems like oily or dry skin. According to results obtained and statistical assessments, the ResNet model and Scattering Networks have the greatest validation accuracy of 98 percent for image classification of oily and dry skin pictures. They have also compared the model with others where scattering networks for SVM have an accuracy of 57 percent, whereas VGG16 gave an accuracy of 95.9 percent on validation sets. And DWT with ANN has an accuracy of 63.2 percent, and Gabor filters and support vector machine have the lowest accuracy of 48.8 percent.

Keywords - Deep Learning, Histogram Equalization, ResNet, Skin Classification, Wavelet Scattering.

1. Introduction

The skin makes up about 15% of the total weight of the human body. A square inch of skin on the human body has approximately 300 sweat glands. Changes in the skin can indicate a medical condition or changes in body health. Skin type varies largely among individuals and can be broadly classified into two types dry skin and oily skin. Knowing what type of skin you can help narrow down the possible skin diseases you might be vulnerable to can also be instrumental in suggesting an effective remedy. The global skin care products market was worth \$140.92 billion in 2020, and it's predicted to grow at a CAGR of 4.69% by 2026. This humongous figure is self-suggestive that people invest a lot in their skin. Normal skin, oily skin, dry skin, combination skin, and sensitive skin are different skin types. The skin of a normal person is neither too dry nor overly greasy. It has very few flaws and is not extremely sensitive. It has a beautiful complexion with barely discernible pores. It has a regular texture and is smooth to the touch. The skin tone is more or less even [1]. Excess oils and pores are related to oily skin produced by hyperactive sebaceous glands [2]. Oily

skin is characterized by a glossy, thick, slippery, and heavy texture. It's more common in younger individuals, but as they become older, their skin becomes drier. Dry skin, commonly known as xerosis, is a common skin ailment that affects people [3]. It is usually not a significant condition, although it might sometimes be difficult to cure. It might be linked to other skin disorders or pathological situations, including diabetes, malnutrition, etc. Dry skin is a troublesome condition that causes itching, scaling, and cracking. It can occur during winter when moisturizing is neglected and might also be caused due to use of harsh skin products. With the increasing awareness about skincare and the increasing number of skin disorders, the need for prompt skin type classification [4] can help pharmacists and dermatologists narrow down to a few from the immense possible scope of potential skin ailments. But most people still don't use the right product according to their skin type as most people are simply unaware of their skin type [5]. So, knowing one's skin type could be very beneficial. This paper proposes a wavelet scattering-based neural network classifier which trains on a small dataset and gives promising results.



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Optimization of quality of service using ECEBA protocol in wireless body area network

Renuka Sahebrao Pawar  & Dhananjay Ramrao Kalbande

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Abstract

Smart healthcare using IoT is the Internet of Healthcare Things (IoHT) that consists of smart monitoring of health remotely by forming the wireless body area networks (WBANs). Each WBAN node having body sensors will collect the periodic medical information of patients and sent to actuators or access points where the medical decision is taken in a multi-hop manner. The IoHT deployment suffers from the two challenges such as cost-effective link establishment and effective access control policies. This paper proposed the novel edge layer protocol called Enhanced Cost-effective and Energy Efficient Based Access control policies (ECEBA). The goal of cost-effective algorithms is to improve the quality of service. The proposed solution is using the lightweight elliptical curve cryptography (ECC) to achieve medial data security and medical user privacy preservation. The ECEBA protocol is implemented and evaluated using NS2. The simulation results prove the efficiency of the proposed protocol over existing methods.

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considered an essential element when reasonably raising Latin heels.

Keywords: Latin dance footwear; biomechanics; electromyography; heel-height.

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Video Analytics based multi-symptoms system for determining progression of Parkinson Disease

by Jignesh Sisodia, Dhananjay Kalbande

Abstract: Parkinson disease is the second most common neurodegenerative disease, and its symptoms tend to increase progressively and affect numerous parts of the body. Parkinson disease patients suffer from symptoms such as rigidity in the body, Bradykinesia, tremors in hands, facial tremor, and freezing of gait. Traditionally assessment of Parkinson disease is based on clinician observation on the severity of symptoms of patients during the visit. Symptoms of Parkinson are highly episodic and cannot be completely observed at the doctors clinic. With the effect of COVID-19, physical visit of the elderly population to the clinic is considered unsafe. Video-based assessment at the patients home led to the solution of avoiding the patient to be exposed to the outside world. We propose a non-invasive video analytics-based assessment of progression of Parkinson disease based on finger tapping and tremor using UPDRS scale. We also propose a video-based technique utilising deep learning and convolutional neural networks which analyse the gait characteristics of patients to identify Parkinson. We intend to distinguish a healthy subject and progression of disease at different stages. These techniques can assist clinical experts for examination of patients to identify the progression of the disease.

Keywords: Parkinson disease; video analytics; deep learning; convolutional neural network.

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Effects of Music Therapy, Mantra Therapy, and Alpha Beta Binaural Beats on Human Energy Fields using RFI Technology

by Karan Sharma, Sudhanshu Choudhary

Abstract: This technical paper is based on research on human energy fields, also known as aura,

Finding Missing Person using AI

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Abstract— Many people and children in the world go missing every day, which the police and local authorities must deal with. The police and municipal authorities deal with many missing persons and kid cases every day. In 2020, National Crime Records Bureau (NCRB) listed around 300 thousand missing persons in India. The general process normally deals with investigation, which requires experience and can be time-consuming if the missing person is migrated. The overall procedure usually entails inquiry, which is time-consuming and requires experience. Our research intends to create an AI system that can identify and evaluate a person's facial features and then compare them to a face in the database, even if they are located years after going missing. Using the Yale Face Dataset on Kaggle containing over 70,000 images for the model, the training of such large dataset improved the model for the Eigenface as well as the GAN algorithm with the use of OpenCV and HaarCascade library helped improve and enhance the accuracy with the result of near 99.

Index Terms— OpenCV, Face Detection, HaarCascade, CNN, Eigenface, GAN.

I. INTRODUCTION

In today's times, there is an increase in the crime rate in the past couple of years. This includes crimes like kidnapping, theft, murders, etc. According to the New York Times report dated 17th April 2022, murders have increased by approximately forty percent since 2018, and other vicious crimes including assaults and murders have also increased drastically. There is a need for help today for a system to support this cause in a time-efficient manner with accurate reports and no scams. Thus, we have developed a system named 'Finding Missing Person using AI' whose main aim is to assist police and better government to tune down lacking humans quickly. Most of the time, the research approach works quite properly, however it's time-consuming and maybe unsuccessful if the missing person has been shifted/moved to an exclusive location. In such cases, the right technique is to undergo CCTV footage and evidence which would again take a lot of time. So, it's very important to fill these gaps by implementing a certain system to benefit society. The facial recognition requires a way to identify or confirm a person's identity using their face [1]. Facial recognition software can identify people in real time as well as in still photos and movies respectively.

Face recognition is a sub-type of biometric security. There may also be a want of automation for automating the mission of locating the specific man or woman via way of means of recognizing a specific photograph and evaluating it with a different photograph to test whether each photograph has the same traits or not [2]. If the missing person is found much later (say 5 years later), then his/her current face will be aged 5 years back by our system using a suitable algorithm checking to the nearest accuracy [3]. By doing this we can come to know whether the lacking man or woman within the photograph clicked from specific vicinity is accurate or not, and if it's far accurate then the police can begin their subsequent steps to locate the man or woman from that area.

AUTIEASE- ANDROID APPLICATION FOR ASD CHILDREN

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ABSTRACT

One of the bigger challenges that the society faces is ensuring that autistic children get education in a proper way, so that they may develop their skills and consequently, contribute to the society. The objective of this study is to design a system which can detect the ASD level and in turn will help autistic children in their education and personality development. The system will be run on the Android platform using a cloud database as a backend, android being the popular platform where a great number of smartphone device applications are run. The system or application will be designed based on studies on the characteristics of children with autism spectrum disorders. The power of augmentative and alternative communication through mobile devices like phones will change the lives of children with autism by helping them to cope with their visual, mental and developmental disabilities. Autism Spectrum Disorder in short ASD is a spectrum of closely related disorders with impairment to some degree in social skills, empathy, communication and flexible behavior leading to restricted activities and interests. Extensive research over the years has proven that the use of android apps with interactive technologies for autism therapy has been quite efficient. The touch screen interface is very appealing and easy to use due to strong visual support compared to the traditional teaching systems. The Detection mechanism will be based on a few questionnaires and will provide analysis based on input provided. The app will design with the learning and evaluation modules by providing all round support to children, parents and tutors.

Keywords: Autistic children, challenges, ASD, Augmentative, Android application

I. INTRODUCTION

Neuro-developmental illness called autism spectrum disorder (ASD) is characterized by issues socially engaging and communicating. From the perspective of the doctor, it can occasionally be challenging to detect Autism. According to Dr. Mark Hyman, autism is a systemic body condition that affects the brain rather than a genetic brain disorder. ASD patients may exhibit a variety of demanding behaviors. Although it can happen later, the most severe type of ASD is usually discovered in the first two years of life. Adults who have this disorder may experience a range of symptoms, from severe ones to small inconveniences. The autism spectrum disorder is one of five pediatric diseases collectively referred to as Pervasive Developmental Disorders (PDD). This is categorized as a challenging neurological disorder. ASD sufferers may encounter a wide range of symptoms. Depending on the symptoms, ASD can range in severity from mild to severe. However, there are a number of conditions that resemble ASD, such as Attention Hyperactivity Disorder (ADHD). A child with ADHD finds it extremely difficult to engage in social interactions. Males are more impacted by ASD than females are. Genetic factors clearly have a significant influence in the onset of this disease, according to research. The real reasons for ASD are yet unknown to science. Asperger's syndrome, a condition related to autism, is characterized by ordinary to above-average intelligence in its sufferers. One's likelihood of acquiring ASD is increased by low birth weight, having an ASD sibling, and having old parents. ASD clinical diagnosis is time- and money-consuming. Medical tests are not used to diagnose it; instead, observations must be made, which takes time. Additionally, a diagnosis might not always be correct. Using machine learning and data analytics, a healthcare professional can more quickly and accurately identify Autism. We have demonstrated accuracy calculations for a number of machine learning methods used to detect autism. Additionally, we created an analytical tool (dashboard) to interactively evaluate and visualize data. In this study, we evaluated a number of publications to determine how machine learning affects the detection of ASD symptoms, how well existing machine learning models perform in this regard, and how well data analysis can reveal trends and patterns. Applications for ASD detection do exist, as we previously noted, but they are not linked to any dashboard. A dashboard makes it simple to clarify one's perspective on ASD.

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considered an essential element when reasonably raising Latin heels.

Keywords: Latin dance footwear; biomechanics; electromyography; heel-height.

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Keywords: Parkinson disease; video analytics; deep learning; convolutional neural network.

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by Karan Sharma, Sudhanshu Choudhary

Abstract: This technical paper is based on research on human energy fields, also known as aura,



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Investigating Abdominal Electrocardiogram (ECG) Signals using Scalogram Analysis for Accurate Characterization

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Abstract

The abdominal electrocardiogram (ECG) has emerged as a promising non-invasive technique for cardiac monitoring. However, accurately characterizing aECG signals remains challenging due to their complex nature and susceptibility to noise. In this study, we investigate the application of Continuous Wavelet Transform (CWT) combined with scalogram analysis for the accurate characterisation of ECG signals. We propose a novel framework that leverages the time-frequency analysis capabilities of CWT and the visual representation provided by scalogram analysis to extract valuable features from ECG signals. Through extensive experimentation on a dataset of ECG recordings, we demonstrate the efficacy of our approach in accurately characterizing various cardiac events, including P-wave detection, QRS complex identification, and T-wave analysis. The proposed method exhibits superior performance compared to traditional time-domain or frequency-domain analysis techniques.

Furthermore, we explore the impact of different wavelet functions and CWT parameters on the characterization accuracy. Our findings highlight the potential of CWT/scalogram analysis as a robust tool for precise and reliable characterization of ECG signals. The outcomes of this research hold great promise for advancing the field of non-invasive cardiac monitoring and have significant implications for improving the diagnosis and treatment of cardiovascular diseases.

Keywords: Abdominal electrocardiogram (ECG) signals, Time-Frequency analysis, Continuous Wavelet Transform (CWT), Scalogram Analysis, P-wave detection, QRS complex identification, T-wave analysis, wavelet functions, cardiovascular diseases.



Automatic question generation: a review of methodologies, datasets, evaluation metrics, and applications

Nikahat Mulla¹ · Prachi Gharpure²

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Abstract

Question generation in natural language has a wide variety of applications. It can be a helpful tool for chatbots for generating interesting questions as also for automating the process of question generation from a piece of text. Most modern-day systems, which are conversational, require question generation ability for identifying the user's needs and serving customers better. Generating questions in natural language is now, a more evolved task, which also includes generating questions for an image or video. In this review, we provide an overview of the research progress in automatic question generation. We also present a comprehensive literature review covering the classification of Question Generation systems by categorizing them into three broad use-cases, namely standalone question generation, visual question generation, and conversational question generation. We next discuss the datasets available for the same for each use-case. We further direct this review towards applications of question generation and discuss the challenges in this field of research.

Keywords Automatic question generation · Natural language generation · Natural language processing

1 Introduction

Automatic question generation (AQG) systems are those in which questions are generated based on a topic or idea or context in natural language from either a paragraph of text or images. Such systems are becoming more popular of late, with their requirement in machine reading comprehension applications, conversational systems, and even educational applications. The eventual goal of AQG systems is the capability to generate questions that are correct syntactically and semantically as well as meaningful in the context of the use-case. For instance, in some cases, the goal is to generate questions on a topic of interest or based on different spans of text in a passage [1]. On the other hand, in conversational systems, say, a question-asking bot, it is imperative to be consistent with the context of the conversation and also at the same time maintain the interest of the user in the conversation [2].

AQG has been widely experimented with, in educational settings. In [3], an attempt is made to generate questions

based on the content of English stories. Questions in five different categories of understanding were framed by extracting syntactic and semantic information present in the stories using natural language processing. Their work assisted specifically to the language learning ability of the learners. The authors compared their generated questions to those that were asked in their collection of book problems and also evaluated them for semantic correctness. In [4], the concept of self-questioning in the context of reading comprehension is explored. In their approach, they generate instructions that help the learners to ask questions relevant to the passage. Children's stories were considered as the dataset of passages. Rather than generating questions randomly, the questions related to the characters' mental states involved in the passages are framed to enable to infer connections between key story characters. Ten different categories of modal verbs were used for constructing questions of three different types (what/why/how) making use of question templates. Evaluation of generated questions was done by testing for acceptability. About 71% of generated questions were marked acceptable, but they suffered from parsing and grammatical errors.

Question Generation Systems fall into one of the following domain categories: closed domain or open domain. In closed-domain question generation, questions are generated

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Leveraging well-formedness and cognitive level classifiers for automatic question generation on Java technical passages using T5 transformer

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Abstract

Automatic question generation is a prevalent area of study for a variety of purposes. Recently, a lot of work has been carried about in this area using deep learning models. One application of such a system is to generate questions about a technical text of interest, which requires additional considerations beyond open-domain question generation. In particular, the datasets used for training deep learning models for question generation are crowdsourced where many samples in the dataset may not be grammatically correct nor have technical accuracy. To devise a question-generation system for technical topics, it is necessary to account for the factors of well-formedness and cognitive levels. To address these issues, we present a hybrid framework in the form of a pipelined architecture that makes use of two additional classifiers before generating questions for text. The first classifier predicts the well-formedness of a question, and the second classifier predicts the cognitive level of questions. We add the two classifiers in a pipeline before passing the questions to a text-to-text transfer transformer (T5) deep learning model for question generation. We trained our framework on the SQuAD Dataset and evaluated several variants of our proposed model using automatic evaluation metrics of BLEU4, ROUGE, and METEOR. Our hybrid model outperformed several other transformer-based models explored in earlier work on these metrics. We further tested the trained framework on a curated collection of technical passages based on the Java programming language. Qualitative assessment using Cohen's kappa for inter-annotator agreement revealed that our proposed pipelined framework, which consists of the well-formedness model, the cognitive model, and the question generator model, can generate more meaningful questions of the target cognitive level as compared to the traditional approach which solely uses the question generator model. Our framework could help generate well-formed questions automatically from the text at different cognitive levels of interest.



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Modelling of FinFET based ON bipolar cell of silicon retina

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Keywords:

Silicon retina
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ABSTRACT

—In this paper, we proposed a SOI FinFET based ON bipolar cell of silicon retina which plays an important role in neuromorphic applications like differential motion detection. Low power consumption along with biological plausibility is one of the crucial parameters of any neuromorphic retinal circuit as in the brain all biological systems consisting of billions of neurons consume power in terms of a few μ watts. First, the output of linkage cell between inner and outer plexiform layer i.e. bipolar cell is reconstructed and verified using developed ionic current based mathematical model which is established considering Hodgkin-Huxley model as the base model, and then biological behavior of the ON bipolar cell is mimicked through circuit using 180 nm, 90 nm and 45 nm Metal Oxide Semiconductor Field Effect Transistor (MOSFET). The power consumption in each case is obtained to be approximately 18.5 nW, 10.5 nW and 1 nW which is much larger than the power consumed by biological bipolar cell. Moreover, the same circuit is implemented with 32 nm Silicon on Insulator Fin Field Effect Transistor (SOI FinFET) device. It is observed that SOI FinFET based ON bipolar cell consumes 17.5 pW power which is less in comparison to MOSFET based bipolar cell. The output of bipolar cell is matched with biological nature of the waveform in each case.

1. Introduction

The organ which allows us to see the world is the eye. Adjustment in sensing of light, focusing and reproduction of the images are the tasks performed by eye. The retina plays a major role in fulfilling the assigned task of vision of the eye. The retina is the multilayer light sensitive structure that comprises of inner plexiform layer and outer plexiform layer. The photoreceptors, horizontal cells and bipolar cells are the basic cells present in outer plexiform layer while bipolar cell, ganglion cells and amacrine cells are the basic cells present in inner plexiform layer. Fig. 1 indicates matching of retinal cells based on the working to the electrical circuits and their output waveforms (S. Barnes et al., 1993), (D. Attwell et al., 1982). The photoreceptor is the light-to-electric signal converter. Horizontal cell filters the signals and ON or OFF bipolar cell gets activated based on comparing the signal available at input. The amacrine cell adds all the signals available from bipolar cell. Thus, the photoreceptor, horizontal cell, bipolar cell and amacrine cell release neurotransmitters when the signal current is greater than the threshold while only ganglion cell generates the action potential. The information in terms of action potential is transmitted to brain via optical nerve.

The function of bipolar cells is to filter the input signal received from photoreceptor in time as well as in space (W. Geoffrey Owen et al., 1989). There are several types of bipolar cells. The subtypes can be identified by observing growth of axon and its connection pattern with other retinal cells. It can also be identified by the response to light stimulus. Depolarization with light occurs for ON type cell while hyperpolarization occurs for OFF type cell. Thus, mainly bipolar cells are divided into ON type cone bipolar cell, OFF type cone bipolar cell and rod bipolar cell. The rod bipolar cells do not have any direct connection with ganglion cells and give only ON type response. The ON bipolar cell is responsible for differential motion detection.

In this paper, section 2 describes the related work done for mimicking the bipolar cell using different approaches. Section 3 first describes the reconstruction of bipolar cell output through developed ionic current based mathematical model and then the implementation of proposed bipolar cell circuit using NMOS and SOI FinFET technologies. Section 4 discusses the obtained outputs and their interpretation and summary in terms of conclusion is mentioned in section 5.

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FinFET based photoreceptor for silicon retina to reduce power consumption

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CMOS
FinFET

ABSTRACT

In this paper, we propose a SOI FinFET device based photoreceptor for silicon retina to reduce power consumption. Nowadays, in the field of neuromorphic engineering the main aim of the researcher is to implement a biological nervous system based on silicon devices. Major performance metrics which influence the design of neuromorphic circuits along with biological plausibility are compactness, low power consumption, scalability and computational complexity. Moreover, to process the information biologically billions of neurons consume only a few watts of power. However, present design of silicon photoreceptor consumes power much larger than actual biological photoreceptor. Hence to achieve low power consumption we are using nanometer SOI FinFET device technology. In this paper, output of the outer retinal cell, namely photoreceptors, is reconstructed and verified with the help of developed ionic current based mathematical model of the retina. The model contains non-linear differential equations based on the fundamental framework of the Hodgkin-Huxley model. In order to closely match the biological response, the selection of parameters of membrane voltage and other membrane dependencies through compartments are done using MATLAB simulation. After mathematical verification, the biological behavior of photoreceptor cell of retinal system is mimicked through circuit using 180 nm Metal Oxide Semiconductor Field Effect Transistor (MOSFET) and 32 nm Silicon on Insulator Fin Field Effect Transistor (SOI FinFET) device. It is observed that SOI FinFET based photoreceptor for silicon retina circuit consumes less power in picowatts matching the power consumption of biological retinal system, as compared with MOSFET based circuits.

1. Introduction

The interpretation of visual information is done biologically with the help of retina. Retina is the light sensitive three-layered structure made up of sensory neurons and complicated neuronal circuits, situated at the back of human eye. The layers in retina are formed with different cells, namely photoreceptors (i.e., rods and cones), horizontal, bipolar, amacrine and ganglion cells. Each cell performs specific task to convert the incident image into electrical signal and helps in understanding it by sending signal to brain via optical nerve.

Since 1990s, researchers started to implement the working of human nervous system using Complementary Metal Oxide Semiconductor (CMOS) technology and later aided through computer systems. They tried to implement vision, auditory and motoring systems of human brain. The human brain interprets the information through analog chemical signals while the computers process digital signals obtained through mathematical modeling of these analog chemical signals

processed using CMOS technology. Neuromorphic computing is a combination of analog and digital systems to mimic the working of human nervous system.

In creation of artificial vision, the CMOS technology has limitation in miniaturization due to atomic limits, in device packing density due to local energy dissipation and in decrement in overall power consumption (Schuller et al., 2015). compared biological neurons and equivalent structure built using silicon. From the comparison of parameters, it has been found that though the speed, size and temperature range is superior in silicon retina compared to biological retina, the major challenge is in reduction of total energy consumption. Another study given by (ScienceDaily, 2021) indicates the energy consumption in human retina is around 12.7 pW vis-à-vis 192 pW in image sensors used in advanced smartphones.

Hence to match the biological processing, improvements are required in nanoscale device design, new functional materials or novel software implementation to achieve desired performance parameters in

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Development of Software Tool to Identify Liquid Maldistribution in Distillation Columns

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Abstract: This paper gives the details of the software developed for the detection of faults in distillation columns using the segmented FFT method. This software tool was tested successfully using simulated data and verified using actual data from a previous study. The software tool correctly detected unequal liquid distribution in sections of the distillation column. This result was helpful for correctly identifying the faults in the distillation columns.

I. INTRODUCTION

A distillation column is an essential item used in the distillation of liquid mixtures to separate the mixture into its component parts, or fractions, based on the differences in volatilities. Our focus would be on packed style distillation columns specifically. A packed column consists of a cylindrical shell containing support plates and a liquid distributor. The cylindrical shell is filled with some sort of packing that rest on the support plate. The packing material offers a large interfacial area for mass transfer. Due to faults in the liquid distributor and beds, there is a high chance of liquid maldistribution which severely affects the quality of the final product. Rectifying such a fault includes opening the column which stops the production that can lead to an enormous loss of revenue to refineries. One of the online methods of identifying faults is Gamma scanning. The data from the scans is traditionally analyzed manually by comparing the graphical plots of the data. A better and faster alternative to this is to use the segmented FFT method.

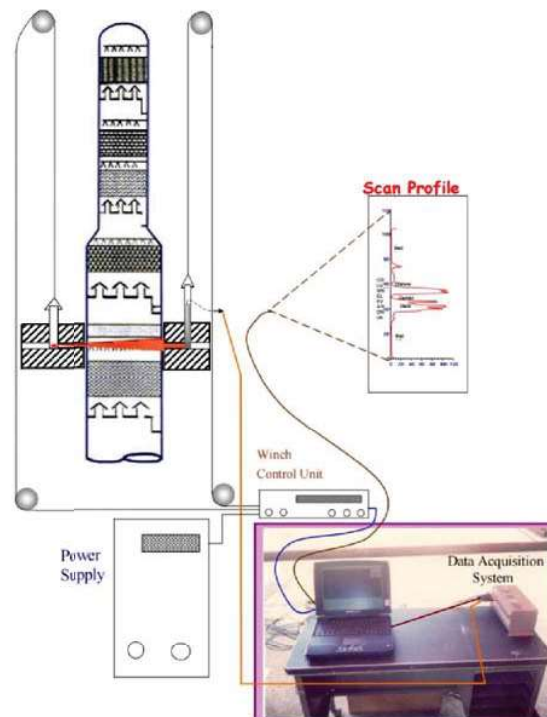


Fig. 1: Setup of the Gamma Column Scanning Equipment [1]

INTERACTIVE ANDROID APPLICATION FOR MARKETING AND PROMOTING SMALL BUSINESSES

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ABSTRACT

A marketing application can give businesses a way to advertise their goods. In the commercial sector, merchants who operate as corporate entities and sell goods and/or services sometimes struggle to attract and retain clients because they are unaware of market pricing and trends. Marketing can be very confusing for small businesses, and it varies greatly depending on the type of business you run. Small business marketing has changed over time and will likely continue to have unanticipated effects on the commercial environment. To attract and keep clients and stand out above the noise, you must constantly come up with new tactics and initiatives. The smart usage of mobile applications is a component of that approach. A business-to-business marketing application provides small-scale businesses with a platform to connect with other business owners and understand market trends.

Keywords: Android Application, Small Scale Business, Digital Marketing, Promotion.

Introduction

In this competitive environment of digital advertising and online businesses, a businessman needs a nice platform which will help him to showcase his vision of the idea. In simple words the platform where he can advertise his business efficiently with the help of online marketing strategies available under one roof rather than using multiple platforms. The platform where the owner can sell at his best and the buyer can purchase with his best.

Business Adda is a platform where all businesses will get to promote their business with other different businesses. This will help small-scale businesses to showcase their good ideas and products with others to increase their sales by great digital marketing. All owners or different business teams can reach out personally to other companies. Also, all the services, products, can be rated by consumers to enhance the quality and their skills.

Not only sellers but also it will be helpful for the buyers!! All the local business owners can buy the products or can access the services provided by the other sellers. One of the best parts of the Business Adda is, the application allows the direct contact between buyers and sellers by avoiding all the third-party resources, to maintain the fluent communication between both the parties.

Literature Review

Halassinos (2022). The aim of this study is to find a market opportunity for members of the IT services association to provide certain online marketing services to small and medium-sized businesses in Prague and the Central Bohemia region. The study examined the use of various online marketing techniques and tools by small and medium-sized businesses and self-employed individuals and classified them based on legal structure, gender of the owner/respondent, and field of business. The analysis revealed that there is a significant difference in how artisans and other businesses use online marketing, across all three surveys, at least at the 5% level of significance. Additionally, the results were consistent for the use of Google Analytics, social networks, and measuring effectiveness on social media.

Dilsad & Murat (2022). In this research, the technology acceptance model was used to investigate how social media marketing and innovation impact the performance of food and beverage companies. A survey was conducted on 163 businesses in Erzurum to collect information within the study's parameters. Structural equation analysis using the AMOS program was used to test hypotheses. The results of the study showed that

Foodie - Online Food Delivery with Customizable Orders

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Abstract— This paper presents a food delivery website with customizable order options that aims to provide customers with a user-friendly platform where they can order food according to their preferences. Customers can customize their orders by selecting specific ingredients and instructions on the website, which has a large selection of menu items. This fixes the common issue of the limited customization options faced by customers with particular dietary requirements or preferences. The website can also be utilised by people who are ill and following a strict food plan recommended by their doctor. The website's user interface is designed to be easy to use and intuitive, and it includes detailed instructions for placing purchases and paying for them. Additionally, the website offers a dependable delivery service that makes sure that food is delivered to customers' doorsteps within the specified window of time. Overall, the food delivery website with customizable order options aims to serve a variety of clients with various dietary requirements and preferences by offering them a convenient and personalized online ordering experience

Index Terms—Online food ordering, customizable orders, dietary preferences, user-friendly platform.

I. INTRODUCTION

With the rise of technology, the food industry has undergone significant changes in recent years. Online food ordering is the process of food delivery or takeout from a local restaurant through a web page or an online Application[1]. Online food ordering has become increasingly popular, and many restaurants and food businesses have adapted to the trend by offering delivery services. The emergence of online food delivery platforms has revolutionized the way food businesses operate, creating new opportunities for growth and expansion in the digital age[2]. However, customers often face limited customization options when ordering food online, which can be frustrating for people with particular dietary preferences or constraints.

To address this issue, a food delivery website with customizable order options has been developed. The website aims to provide customers with a user-friendly platform where they can order food according to their preferences. The website offers a vast selection of menu items. Customers can make their own customized orders by specifying ingredients and instructions for the same.

The website resolves the frequent problem of the restricted customization possibilities faced by consumers with certain dietary restrictions or preferences.

Customers with special dietary requirements, such as those who are vegan, vegetarian, gluten-free, or have food allergies, can tailor their orders to suit their requirements. Additionally, customers can choose to have their orders customized based on their preferred ingredients or cooking techniques. The website makes sure that all consumers can enjoy their meals without compromising on their preferences or health issues by offering these personalized possibilities.

Apart from catering to dietary restrictions and preferences, online food delivery with customizable orders has the potential to serve another demographic: individuals who are ill and on a strict diet prescribed by their doctor. Frequently, these people are unable to prepare their own meals because they live alone. They can upload their recipes and place orders online using customizable orders, and the restaurant will fulfil them according to the details they supply. Ensuring they receive meals that meet their dietary needs without having to spend energy preparing, this can significantly improve their quality of life.

The user interface of the website is made to be simple and intuitive, with clear instructions for placing orders and making payments. Additionally, the website provides a reliable delivery service, ensuring that food is delivered to customers' doorsteps within the promised time frame.

Overall, the food delivery website with customizable order options aims to provide customers with a personalized and convenient online ordering experience. By offering a vast selection of menu items and customizable order options, the website hopes to cater to a wide range of customers with different dietary needs and preferences.

II. LITERATURE REVIEW

An overview of the technologies now in use in the field of online food delivery is provided in this section -

Rising labour costs provide a big challenge to firms in the food industry. Some advocate replacing human labour with contemporary technology to cut costs. The suggested remedy is an online system for placing meal orders that are intended for fast food establishments, takeaway, or college cafeterias. The system can automate the ordering procedure for the client as well as the restaurant, making it simpler and maybe



INTERVENTION RECOMMENDATION WEB APPLICATION BASED ON USER INTERESTS FOR MENTAL WELL-BEING

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Abstract: The sensitive topic of 'mental health' is increasingly gaining more recognition and respect among people of all age groups. Starting from children to senior citizens, everyone needs aid that will help them keep their mental state healthy. The seemingly fast-paced world and rising competition have contributed to disturbing one's mental condition to an extent. The Coronavirus pandemic has accelerated the decline in the mental health of the majority of people. In such a condition, it becomes both - necessary and beneficial - to use the power of technology in helping people maintain a positive state of mind. With the vast range of advantages that machine learning offers, it becomes easy and effective to cater to individuals as per their choices and preferences. Machine learning is used widely in the healthcare sector because of the benefits it provides by equipping the ability to train models to perform specific tasks based on the information supplied. Similarly, machine learning can be used to help people improve their mental state by learning their favorite activities to engage in to feel better and generate recommendations for activities accordingly. In such a way, it can help uplift a person's mood by giving him a range of options for tasks to perform based on his liking. In an attempt to help people of all age groups improve their mental health, an application was developed which takes into consideration the user's likings and current mental state and recommends intervening activities to participate in.

Index Terms - mood, mental health, social media, Gmail, chrome extension, deep learning, LSTM, recommendation, hobby, intervention, activity.

I. INTRODUCTION

The World Health Organization (WHO) announced in September 2021 [1] that depression affects around 280 million people worldwide. Aside from depression, there are numerous other mental disorders that affect people's daily lives. Since 2012, the percentage of youth depression has risen from 5.9 to 8.2 percent [2]. Depression symptoms can have an impact on academic performance as well as personal relationships.

Mental illness affects people of all ages, but teenagers are the most vulnerable demographic. Mental illness affects or will impact 1 in 5 teenagers at any given point in their teenage life [3]. The majority of youngsters are uncomfortable sharing or expressing their emotions in front of their family members. People who work or own businesses frequently face enormous stress, which, if not managed effectively, can lead to major emotional problems. Senior persons frequently suffer feelings of loneliness, anger, aggravation, emptiness, and other negative emotions that, if unaddressed, can contribute to mental illness.



L-MM Auger electron emission from chlorinated organic molecules under proton impact: Angular distribution and total cross section measurement

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Highlights

- Angle and energy differential cross sections for chloromethanes.
- L-MM Auger emission cross section for proton impact energy 125 keV to 275 keV.
- Absolute total cross section and isotropic angular distribution measurements.

Abstract

We have measured the absolute cross section for L-MM Auger electron emission in chlorinated methanes and benzene chloride molecules in collision with keV energy protons. The measurements have been performed with four different chlorine containing organic molecules viz CCl₄, CHCl₃, CH₂Cl₂ and C₆H₅Cl. We have measured the angular distribution of chlorine L-MM Auger electrons at backward emission angles (90° – 150°). The angular distribution shows an isotropic character. We also studied the projectile energy dependence of the total L-MM Auger yield for proton energy ranging from 125 keV to 275 keV. Carbon K-LL Auger yield was also obtained in the same experiments.